

Business Analysis Case Study – Inventory Management System (IMS)

Introduction

Retail businesses operate across multiple locations and handle thousands of products on a daily basis. Efficient inventory management becomes challenging when stock movements are tracked manually or when retail stores, warehouse operations, and sales systems are not interconnected. It often leads to stock mismatches, delays in replenishment, and poor visibility of available products across different stores.

To solve these issues, a software development company has initiated a project to build a Inventory Management System (IMS) for a retail client. As the Business Analyst for this project, my responsibility is to gather requirements, identify business needs, document processes, analyse stakeholder expectations, and assist in designing a solution that ensures real-time inventory tracking, automated replenishment, and improved reporting.

This assignment covers the complete business analysis lifecycle—stakeholder identification, requirement gathering, BRD preparation, user stories, use cases, wireframes, test cases, acceptance criteria, communication plan and timeline. The goal is to define a structured and scalable IMS solution that meets the client's operational needs efficiently while reducing manual effort, improving accuracy, and enhancing stock visibility across multiple outlets.

1. Stakeholder Analysis

1.1 Stakeholder Identification Table

Stakeholder	Role in the Project	Why They Matter
Business Owner / Client	Project sponsor & funding provider	Approves budget, requirements, and final project delivery
IT / Technical Team	Maintains system infrastructure	Provides servers, database, and integration support
Software Development Team	Builds the IMS features	Needs well-defined requirements & workflows to develop
QA & Test Engineers	Test the system before release	Ensure quality, detect bugs, evaluate performance
Store Managers	End-users operating inventory in stores	Track product quantity, respond to low-stock alerts, request replenishment
Warehouse Team	Handles stock storage & movement	Updates incoming/outgoing stock records regularly
Sales Team	Processes customer purchases	Requires real-time automatic stock deduction during billing
Project Manager	Oversees execution and task coordination	Ensures timely delivery, milestone tracking, and stakeholder alignment
Customers (Indirect)	Purchasers of retail items	Benefit through product availability & reduced out-of-stock events

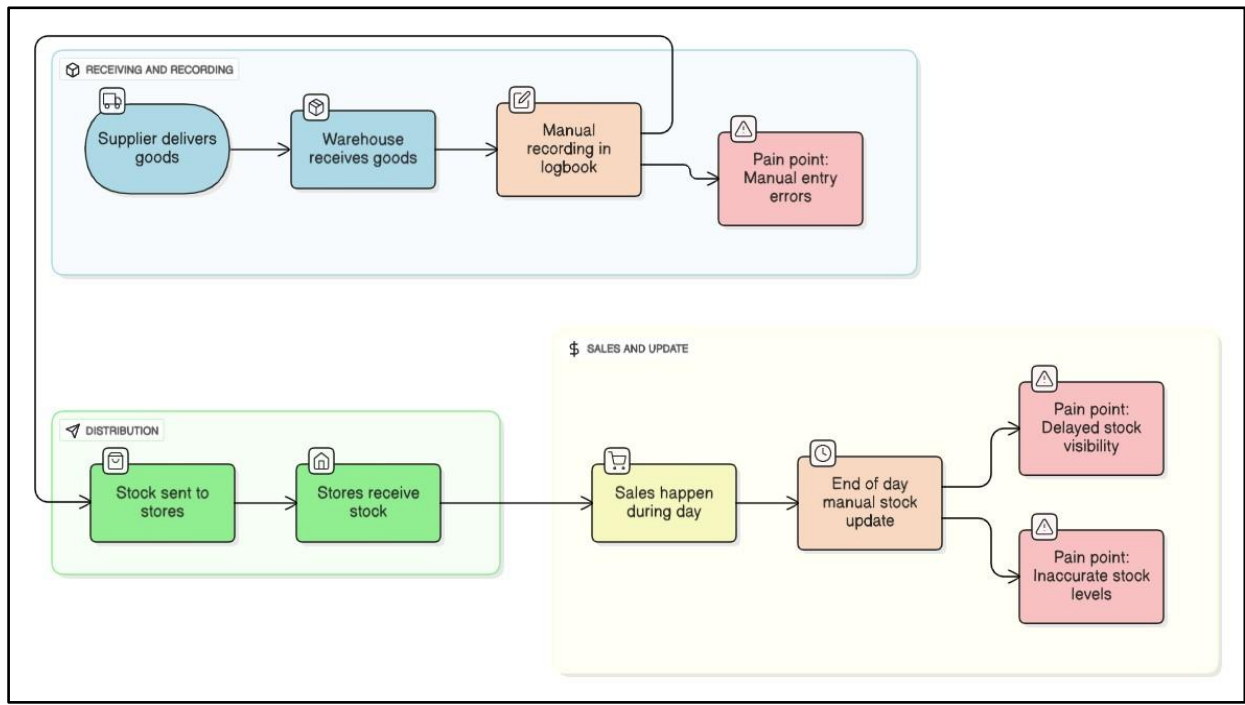
1.2 Stakeholder Communication Plan

A Business Analyst must ensure everyone is informed properly — so we create a **communication matrix**.

Stakeholder Group	Information Shared	Method of Communication	Frequency
Business Owner / CEO	Project status, risks, milestone reports	Email + Review Meetings	Weekly
Project Manager	Work progress, requirement updates	Stand-up Meetings	Daily
Development Team	Requirements clarification, user stories	Slack/Teams + Sprint Meetings	Daily– Alternate Days
QA/Testers	Test coverage & bug status	Test Review Meeting	Weekly + Pre-Release
Store & Warehouse Managers	UI review, workflow validation	Feedback Calls / Demo Sessions	Bi-Weekly
Entire Stakeholder Group	Final review, change approvals	Presentation + Email Summary	At Major Milestones

2. Business Process Mapping & Requirements Gathering

2.1 Current Inventory Process



Pain Points in the Current System

Pain Point	Description
Manual Entry Errors	Stock is recorded manually which increases chances of incorrect data entry, missed updates, and human mistakes.
Delayed Stock Visibility	Stock levels are not updated in real time, so stores cannot immediately know how much inventory is available.
Inaccurate Stock Levels	Due to delayed and manual updates, the recorded stock quantity often does not match the actual physical stock, leading to confusion and stock-outs.

These problems clearly show why the company needs a centralized **Inventory Management System (IMS)**.

2.2 Business Requirements Document (BRD)

A. Business Goals

The Inventory Management System aims to:

- Provide real-time stock visibility across all stores and warehouse units
- Automate stock updates to avoid human dependency
- Trigger low-stock alerts and support quick replenishment
- Improve decision-making using accurate inventory data & report analytics
- Reduce losses caused by stock-outs, overstocking or mismanagement

B. Project Scope

In-Scope

- Real-time stock tracking & updates
- Low-stock alert & automatic reorder suggestions
- Integration with sales system for automatic stock deduction
- Inventory dashboard with KPIs & analytics
- Store-to-store stock transfer functionality
- Inventory reporting (daily/weekly/monthly)

Out-of-Scope

- Mobile application support (future phase)
- Supplier management automation
- Accounting/finance system integration

C. Functional Requirements

#	The System Must Be Able To...
1	Display real-time stock quantity across all stores
2	Auto-trigger alerts when items fall below threshold
3	Deduct stock automatically after sales billing
4	Allow manual update/edit of stock quantities
5	Support barcode scanning for product entry
6	Generate and download reports (PDF/CSV)
7	Maintain history of all stock transactions
8	Provide dashboard view with analytics KPIs
9	Allow role-based access and permissions
10	Enable store-to-store transfer management
11	Suggest reorder quantity using sales trends
12	Support bulk upload of stock via CSV/Excel

E. Assumptions & Constraints

Assumptions	Constraints
Users will be trained before system rollout	Limited budget and fixed delivery deadline
Internet connection available across stores	Dependency on API integration availability

Hardware scanners will be provided	Restricted workforce for development timeline
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3. User Stories & Use Cases

3.1 User Stories

User Story	Story
1	As a store manager, I want to view real-time inventory levels so that I can prevent stock-outs.
2	As a warehouse assistant, I want to update received stock quickly so that quantity stays accurate.
3	As a sales executive, I want stock to auto-deduct after billing so that I don't have to update manually.
4	As a business owner, I want graphical inventory reports so that I can analyse store performance easily.
5	As a system admin, I want to manage user access roles so that sensitive data remains secure.

3.2 Use Cases (Two System-Based Interactions)

Use Case 1 — Low Stock Alert Generation

Field	Description
Use Case Name	Low Stock Alert
Actors	Store Manager / System
Pre-Conditions	Product stock exists in the database; minimum threshold defined
Main Flow	1. System checks product quantity regularly 2. Quantity drops below threshold 3. System generates low-stock alert 4. Store manager receives notification 5. Replenishment action can be triggered
Alternate Flow	If auto-replenishment is enabled, system may auto-create purchase order without manual alert
Post-Condition	Store manager becomes aware of shortage & avoids stock-outs

Use Case 2 — Automatic Stock Replenishment

Field	Description
Use Case Name	Auto Replenishment
Actors	Warehouse Staff / System
Pre-Conditions	Low-stock alert identified; stock transfer rules defined
Main Flow	1. System generates replenishment request 2. Warehouse staff reviews and approves 3. Stock quantity deducted from warehouse 4. Stock delivered/assigned to store 5. System updates both store & warehouse stock automatically
Alternate Flow	If warehouse is out of stock → request is marked pending/forwarded to supplier

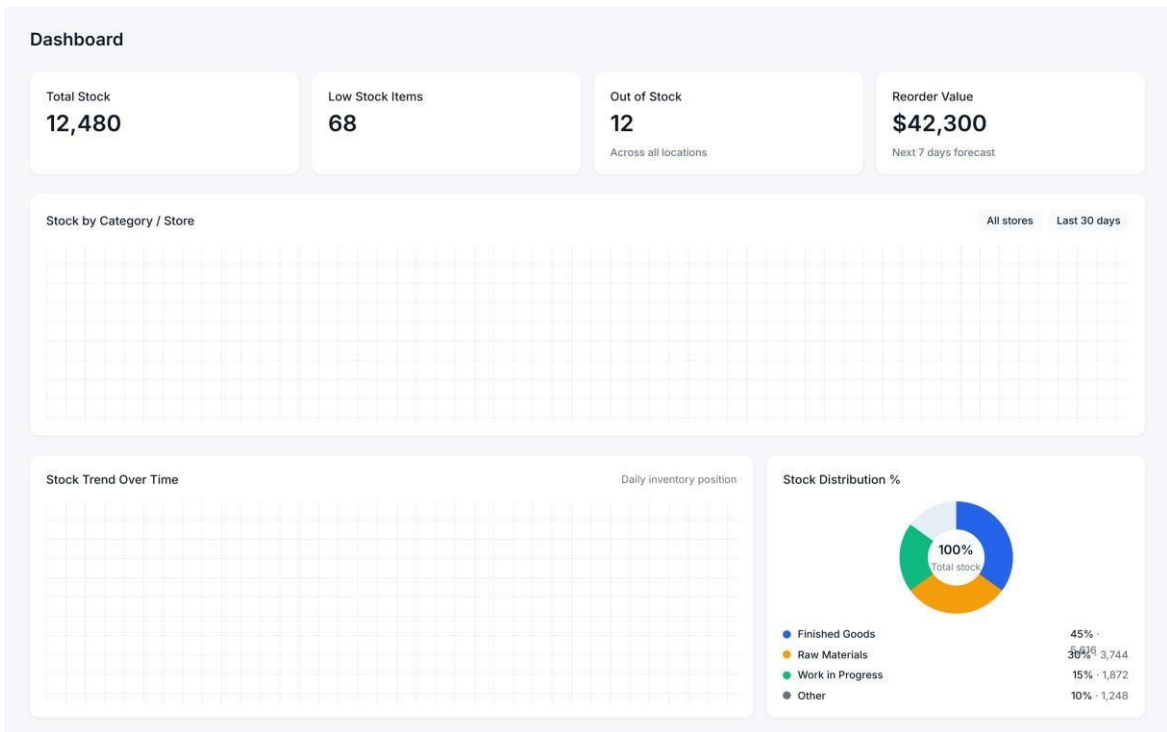
Post-Condition	Store receives product before shortage impacts sale
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4. Functional Prototyping & Wireframes

4.1 Wireframe 1 — Inventory Dashboard (Home Screen)

This is the first screen a user sees after login. It gives a quick overview of real-time inventory, alerts, and overall stock performance.

Wireframe Layout Representation



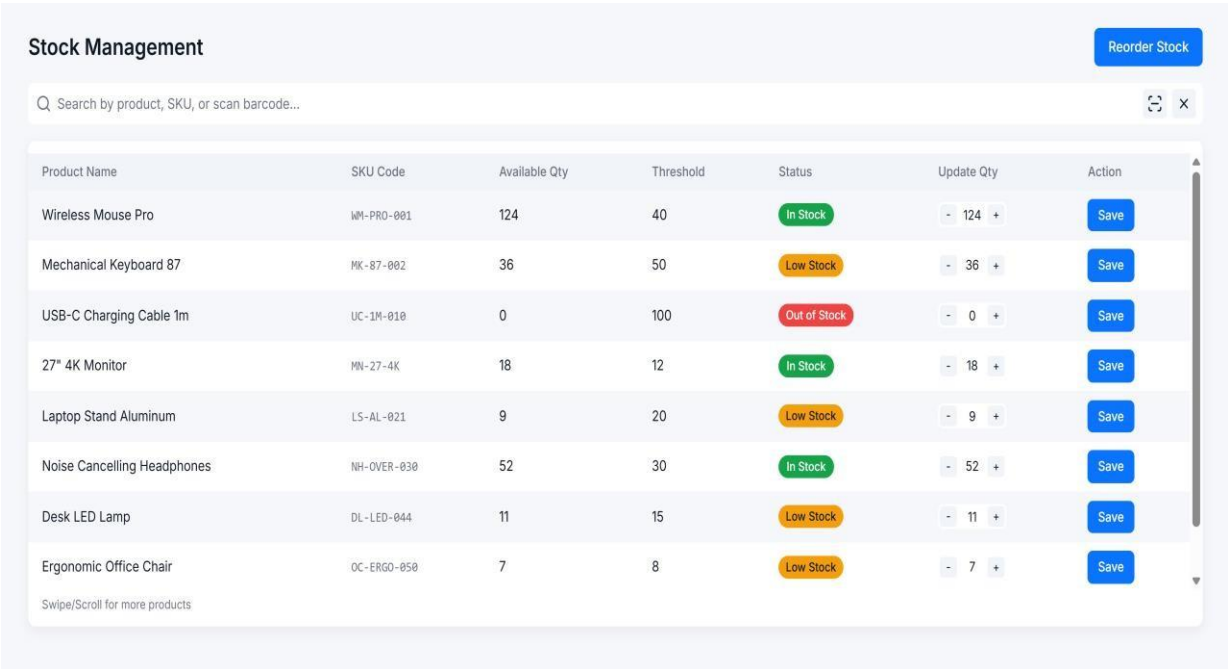
Why this wireframe is effective

Benefit	Explanation
Quick decisionmaking	KPIs and charts allow users to understand inventory health instantly
Better visibility	Real-time stats reduce stock-out chances
Visual insights	Trends and category breakdown help in planning and forecasting

4.2 Wireframe 2 — Stock Management Interface

This screen is used to **update stock, scan products, adjust quantity, view item details, and reorder stock.**

Wireframe Representation



Why this UI helps users

Feature	Benefit
Search + scan input	Allows fast product lookup and update
Editable table	Helps staff correct stock instantly
Reorder function	Prevents stock-outs automatically
Status view	Shows Low/OK/Out-of-stock at a glance

5. Test Cases & Acceptance Criteria

This section is used to check whether the Inventory Management System is working as expected.

We will prepare **3 test cases** based on functional requirements and then define **Acceptance Criteria**, which means what conditions must be fulfilled for the system to be considered successfully completed.

5.1 Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Pass/Fail Criteria
TC-01	Verify low stock alert generation	1. Select an item 2. Reduce quantity below threshold 3. Save changes	System should automatically trigger a lowstock alert	Alert is generated and visible to store manager
TC-02	Manual stock update through stock management interface	1. Search SKU / scan barcode 2. Edit quantity field 3. Click 'Save'	New stock value must reflect immediately in the system	Stock is updated without delay (within 2–3 seconds)
TC-03	Automatic stock deduction after sale	1. Process a sale from POS 2. Complete billing 3. Check inventory screen afterwards	Quantity should reduce according to sold units	Stock deduction matches sale quantity exactly

These cover **major workflow + critical functionality**, ensuring the system works smoothly in real operations.

5.2 Acceptance Criteria (Completion Validation)

The project will be considered successful when:

Criteria	Acceptance Condition
Real-time Inventory Sync	Stock reflects changes within 2–3 seconds after update/sale
Automated Alerts	Low-stock alert is triggered at threshold quantity
Error-Free Calculations	Inventory matches actual physical quantity
User-Friendly Interface	Staff can operate system with minimal training
Reporting	Dashboard and reports generate successfully in one click
Role Permissions	Different users access only assigned features
End-to-End Flow Validation	Warehouse → Store → Sale → Report works smoothly
UAT Sign-Off	Final approval received from both store & warehouse users

6. Communication Plan & Project Timeline

6.1 Communication Plan

A Business Analyst must ensure that all stakeholders receive timely updates regarding project status, risks, and decisions.

Below is the communication plan for the IMS Project:

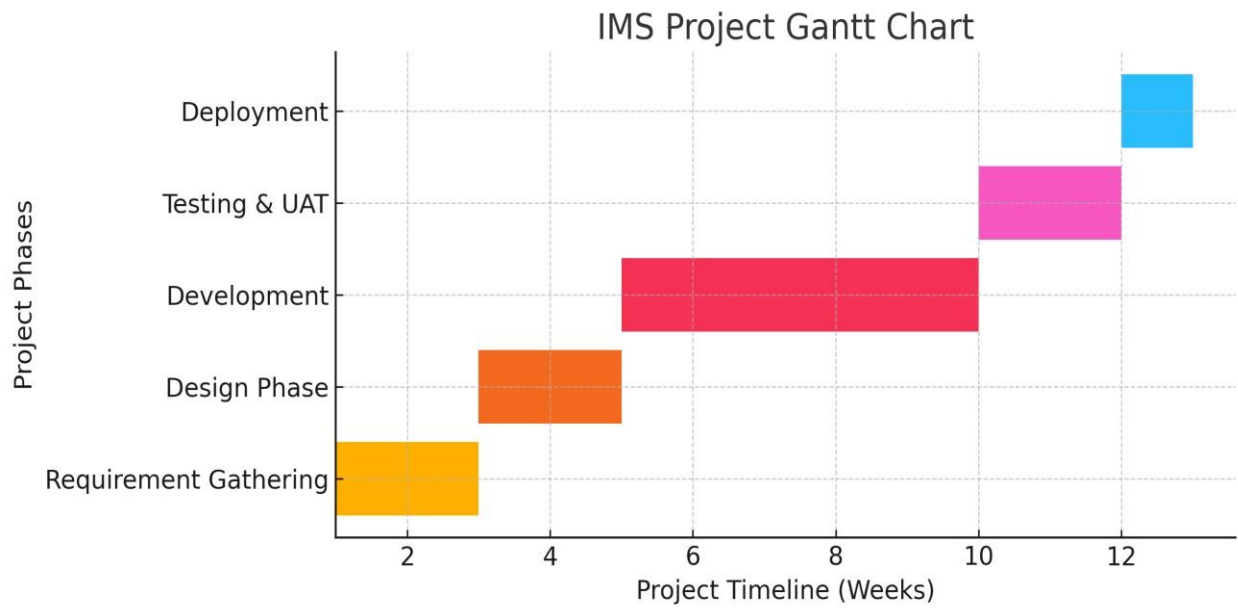
Stakeholder Group	Communication Type	Method	Frequency
Business Owner / Client	Progress Review, Risk Updates	Email + Zoom Meetings	Weekly
Project Manager	Sprint Progress, Issues	Daily Stand-ups + Reports	Daily
Development Team	Requirement Clarification + Tasks	Scrum Meetings / Jira Boards	Daily / Alternate Days
QA Testers	Bug Reports, Testing Feedback	Test Logs + Review Meetings	Weekly & Pre-Release
Store Managers	Feature Demo, User Feedback	Video Meetings + Demo Sessions	Bi-Weekly
Warehouse Team	Training & Usage Support	Live Walkthrough Sessions	After Deployment
Full Team	Final Release Discussion	Presentation + Closure Report	At Project Completion

This ensures **smooth collaboration, short feedback loops & minimum delay in execution.**

6.2 Project Timeline (12-Week Delivery Plan)

Phase	Duration	Timeline
Requirement Gathering + Analysis	Week 1-2	BRD + Stakeholder Confirmation
Wireframes + UI/UX Design	Week 3-4	Dashboard & Stock Module Screens
Development Phase	Week 5-9	Core IMS Features + Integrations
Testing + UAT	Week 10-11	Bug Fixing, Performance Check
Final Deployment & Handover	Week 12	Live Rollout + Training

6.3 Gantt Chart



Represents work in **progress timeline**

This visual chart clearly shows work distribution **week-by-week**.